Ambient Smart Homes

Making Products for Healthy Homes

Market Opportunity

Most smart home products focus on two areas: managing temperature and automating appliances, overlooking the most important function of our homes.

Providing a healthy and safe living space depends significantly on indoor air quality.

Current alternatives fall short in three main ways:

1. **Accuracy**: measure a single point at a time
2. **Convenience**: require an electrical outlet or frequent charging
3. **Insights**: do not provide a way to improve air quality

Smart Home Market:

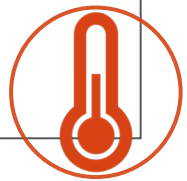
1. McKinsey: 25% of homes (~30 million) in US are 'connected'
2. Statista: current annual revenue of the Smart Home market in the US is $27 billion

Proposed Solution

Ambient plans to make a product that allows everyone to improve health of their home by monitoring and adjusting three fundamental air quality parameters.

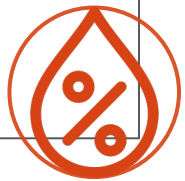
- Impacts comfort and productivity
- Correlated with humidity
- Energy intensive

Temperature



- Must be within a specific healthy range
- Correlated with temperature
- Impacts spread of contaminants

Humidity



- Organic: pollen, mold, bacteria
- Inorganic: VOCs, SVOCs, Ozone (O_3)
- Other airborne pathogens

Contaminants



3-Phase Product Roadmap

1 Monitoring

Ambient monitors 3 key parameters through a collection of sensors

1. Temperature
2. Humidity
3. Contaminants



Sensors : Monitoring

2 Recommendations

Ambient provides data-driven recommendations across 5 categories

1. Change temp
2. Change humidity
3. Increase airflow
4. Turn-on purification system
5. Replace materials



Sensors : Monitoring Recommendations

3 Automation

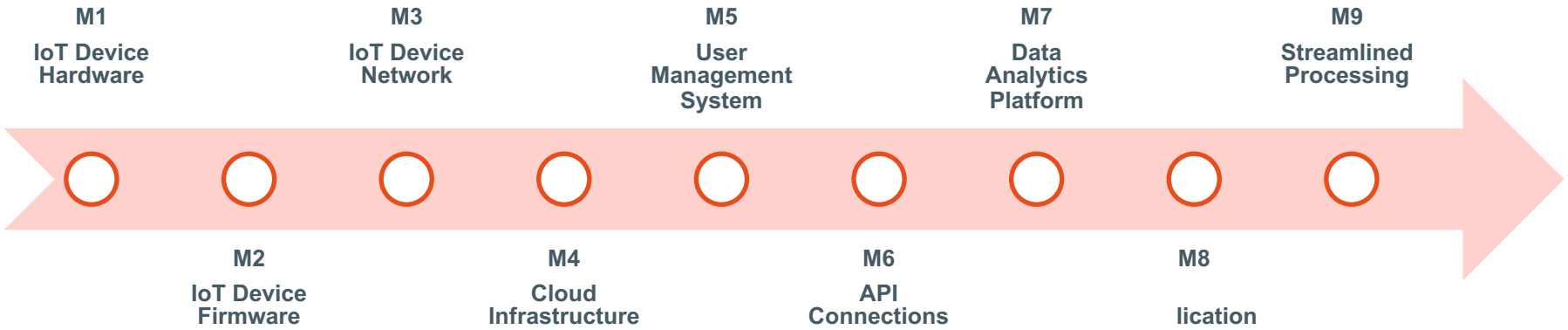
Ambient manages 4 groups of devices to automatically improve air quality

1. Thermometers
2. Humidifiers
3. Purifiers
4. Fans and Ventilators



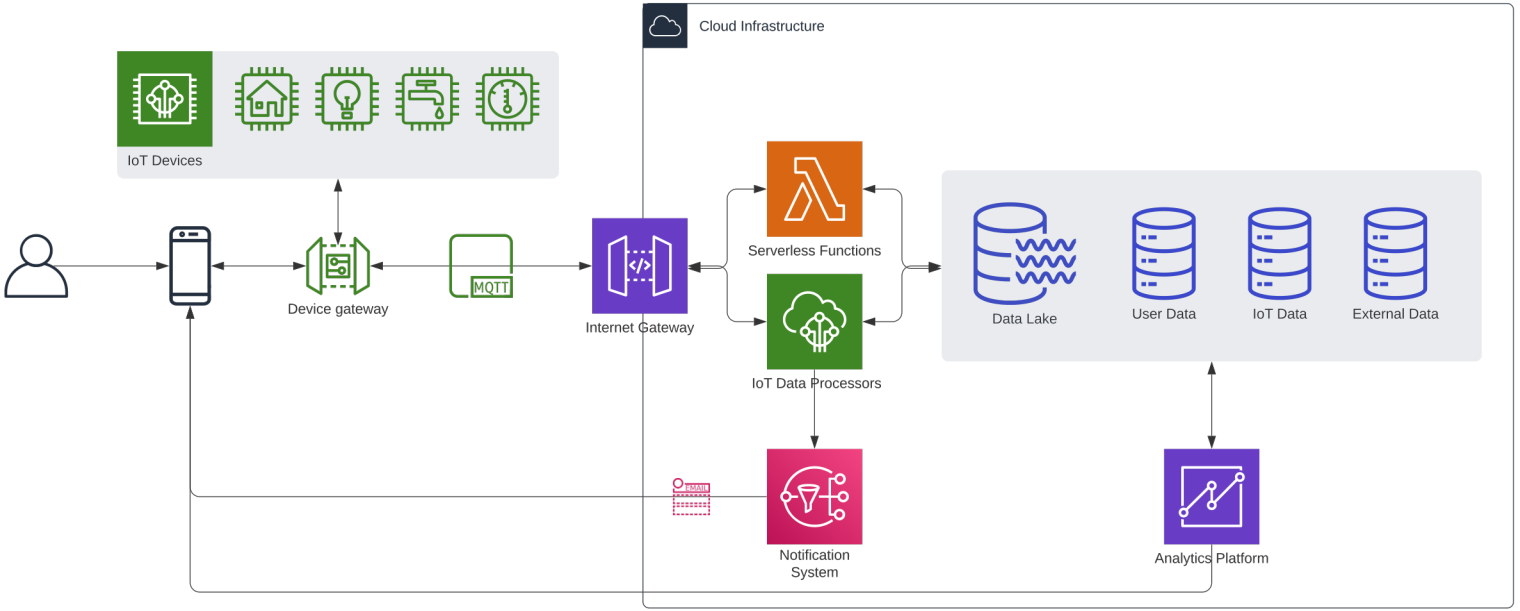
Sensors : Monitoring Recommendations Automation Actuators

The process for building the minimum viable product (MVP) consists of nine main steps and takes approximately two to three months, depending on hardware procurement time.



Conceptual Design

The system architecture consists of four interconnected layers: IoT devices, communication network, , and cloud infrastructure.

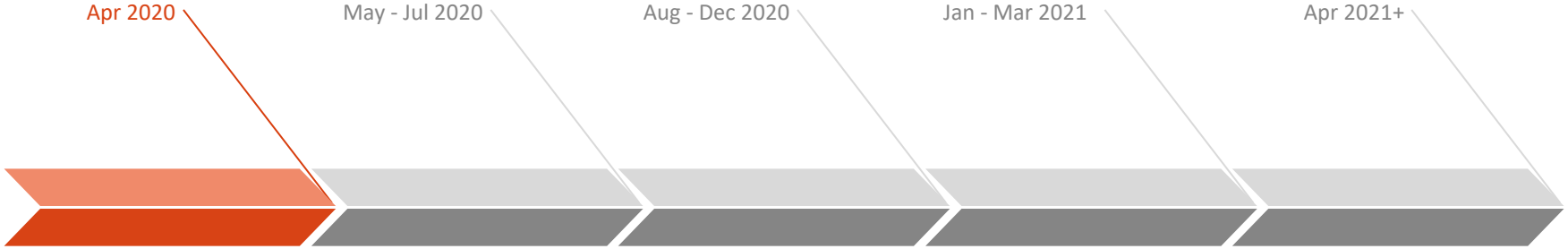


Innovation

The core innovation behind the product is the system design and technology integration, however, there are several opportunities to leverage cutting-edge research.

Layer	Component	IP Opportunity	Advantage
IoT Device	Device hardware	Highly accurate and low-cost sensors	purchase cost • Improve system accuracy
IoT Device	Device hardware	Energy efficient electronic components	operational costs • Increase convenience by charging less often
	Device firmware	New embedded system approaches for processing data at the edge	• Real-time data processing • Lighten load on the cloud
Network Connectivity	Gateway design	A streamlined way to connect IoT devices to the cloud	hardware costs • Improve system performance
Multi-layer	lication	Analytics for accurately evaluating multiple sources of data to provide tailored recommendations through the	the need for automation • Empower users to improve air quality independently

Timeline



Planning

Establish a preliminary hypothesis about the market and create a conceptual design.

Discovery

Identify the primary market, establish a network of partners and advisors, and create a prototype development plan.

GRIP Hothouse

Confirm the IP, build the prototype, and create a preliminary business model.

Testing

Test the Ambient platform with a select group of users and partners.

Roll-out

Confirm anchor partners, build a strong team, and roll-out first version of the product.

Thank You



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- 5 years in product and project management roles with international tech startups
- Experienced in design of engineering systems and digital technology integration
- Led multi-disciplinary and cross-cultural teams in North America and Asia
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